rapid process, generally taking a tiny fraction of a second to happen. From a human perspective, the wave function branches essentially instantaneously (although that's just an approximation). So the branching happens first, and we only find out about it slightly later, for example, by looking to see whether the electron went up or down when it passed through the magnetic field.

For a brief while, then, there are two copies of you, and those two copies are precisely identical. Each of them lives on a distinct branch of the wave function, but neither of them knows which one it is on.

You can see where this is going. There is nothing unknown about the wave function of the universe—it contains two branches, and we know the amplitude associated with each of them. But there is something that the actual people on these branches don't know: which branch they're on. This state of affairs, first emphasized in the quantum context by physicist Lev Vaidman, is called *self-locating uncertainty*—you know everything there is to know about the universe, except where you are within it.

That ignorance gives us an opening to talk about probabilities. In that moment after branching, both copies of you are subject to self-locating uncertainty, since they don't know which branch they're on. What they can do is assign a credence to being on one branch or the other.

What should that credence be? There are two plausible ways to go. One is that we can use the structure of quantum mechanics itself to pick out a preferred set of credences that rational observers should assign to being on various branches. If you're willing to accept that, the credences you'll end up assigning are exactly those you would get from the Born rule. The fact that the probability of a quantum measurement outcome is given by the wave function squared is just what we would expect if that probability arose from credences assigned in conditions of self-locating uncertainty. (And if you're willing to accept that and don't want to be bothered with the details, you're welcome to skip the rest of this chapter.)

But there's another school of thought, which basically denies that it makes sense to assign any definite credences at all. I can come up with all sorts of wacky rules for calculating probabilities for being on one branch of the wave function or another. Maybe I assign higher probability to being on a branch where I'm happier, or where spins are always pointing up. Philosopher David Albert has (just to highlight the arbitrariness, not because he thinks it's reasonable) suggested a "fatness measure," where the